

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

PREPARED BY			
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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC
1.1	3/19/24	Fill the bottle only to the base of the neck to reduce overflowing and corrosion of the BOD stirrer.	TAC

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

Table of Contents

1. Title	3
2. Applicability	3
3. Purpose	3
4. Method	3
5. Health & Safety	3
6. General Precautions	3
7. Training	4
8. Required Materials	4
9. Procedure	5
9.1 Reagent Preparation	5
9.2 Sample Collection	7
9.3 Sample Analysis	8
9.4 Sample Analysis	9
10. Documentation	9
11. Cleaning and Maintenance	9
12. Troubleshooting	10
13. QA/QC	10
14. References	12

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

1. Title

Standard Operating Procedure for Biochemical Oxygen Demand (BOD) Determination

2. Applicability

This SOP provides an indirect measure of organic matter as it measures the Biochemical Oxygen Demand (BOD) of microorganisms in seawater from shallow (<30 m) coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze BOD at surface and bottom seawater from shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef BOD data following the Quality Assurance Project Plan.

4. Method

Standard Method 5210 B (Baird et al., 2017).

5. Health & Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.

6. General Precautions

- 6.1 If analysis begins within 2 h of collection, cold storage is unnecessary.
 - 6.1.1 Otherwise, keep samples at 6°C between collection and analysis.
- 6.2 Begin analysis within 6 hours and no later than 48 hours from time of collection.
- 6.3 Ensure the BOD meter remains plugged in and powered to maintain probe polarization.
- 6.4 Clean bottles with a detergent, rinse thoroughly, and drain them before use.
- 6.5 Discard any reagents with signs of precipitation or biological growth on the bottles.
 - 6.4.1. Use reagent grade or better for all required chemicals.
 - 6.4.2 Use Millipore reagent-grade water to make all solutions.
- 6.6 Control temperature at 20 ± 1 C° during incubation.
 - 6.5.1 Exclude all light to avoid production of Dissolved Oxygen (DO).

7. Training

- 7.1 Personnel shall receive training on how to complete the corresponding forms for this SOP.
- 7.2 Personnel shall be trained in the proper use of the Orion pH meter and the BOD probe

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

8. Required Materials

8.1 Materials

- 8.1.1 Incubation bottles: DWK Life Sciences Wheaton 300mL Black Borosilicate BOD Bottle
- 8.1.2 1 L polypropylene bottle for sample collection
- 8.1.3 Reagent storage bottles: 1 L glass bottles
- 8.1.4 Pipette tips (1 mL)
- 8.1.5 Kimwipes
- 8.1.6 Millipore Ultra Pure Water
- 8.1.7 Onset HOBO Temperature Logger
- 8.1.8 Aluminum foil

8.2 Equipment

- 8.2.1 Microbiological Incubator, 180 L, Stainless Steel
- 8.2.2 Thermo Scientific Orion Auto Star A213 benchtop RDO/DO meter
- 8.2.3 Thermo Scientific Orion AUTO-STIR BOD Probe
- 8.2.4 Analytical Balance

8.3 Required Chemicals

Compound	CAS No.	Safety classification	
Monopotassium phosphate KH_2PO_4	7778-77-0	----	Not a hazardous substance
Dipotassium phosphate K_2HPO_4	7758-11-4	GHS05, GHS07	Warning/Danger - Corrosive, Irritant
Disodium phosphate $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$	7558-79-4	GHS05, GHS07	Warning/Danger - Corrosive, Irritant
Ammonium chloride NH_4Cl	12125-02-9	GHS07	Warning - Irritant
Magnesium sulfate heptahydrate $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	10034-99-8	----	Not a hazardous substance
Calcium chloride (CaCl_2)	10043-52-4	GHS05	Warning/Danger - Corrosive
Iron (III) chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$)	10025-77-1	GHS05	Warning/Danger - Corrosive
Glutamic acid	56-86-0	—	Not a hazardous substance
Glucose	50-99-7	—	Not a hazardous substance
Distilled or reagent-grade water		----	

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

8.4 Reagents Standards, and Preparation Instructions

8.4.1 Phosphate buffer solution

Reagent	Amount	Instructions
KH_2PO_4	8.5 g	1. Dissolve 8.5 g KH_2PO_4 , 21.75 g K_2HPO_4 , 33.4 g $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$, and 1.7 g NH_4Cl into ~500 mL Millipore water 2. Dilute to 1 L 3. Check that the resulting pH is 7.2 4. Storage indefinitely at 20°C unless precipitation or biological activity is observed in solution.
K_2HPO_4	21.75 g	
$\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$	33.4 g	
NH_4Cl	1.7 g	
Millipore water	1 L	

8.4.2 Magnesium sulfate (MgSO_4) solution

Reagent	Amount	Instructions
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	22.5 gr	1. Dissolve 22.5 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in Millipore water 2. Dilute to 1 L 3. Storage indefinitely at 20°C unless precipitation or biological activity is observed in solution.
Millipore water	1 L	

8.4.3 Calcium chloride solution

Reagent	Amount	Instructions
Calcium chloride (CaCl_2)	27.5 gr	1. Dissolve 27.5 g CaCl_2 in Millipore water. 2. Dilute to 1 L. 3. Storage indefinitely at 20°C unless precipitation or biological activity is observed in solution.
Millipore water	1 L	

8.4.4. Ferric chloride solution

Reagent	Amount	Instructions
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	0.25 gr	1. Dissolve 0.25 g $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ in Millipore water and 2. Dilute to 1 L 3. Storage indefinitely at 20°C unless precipitation or biological activity is observed in solution.
Millipore water	1 L	

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

8.4.5 Glucose Glutamic Acid Solution

Reagent	Amount	Instructions
Glutamic Acid	196.0 mg	1. Measure 500 mL Millipore water in a graduated cylinder and add to a 500 mL Pyrex storage bottle. 2. Weigh 196.0 mg glutamic acid and add to the water. 3. Weigh 196.0 mg glucose and add to the water. 4. Shake well to ensure the solution is fully mixed. 4. Storage indefinitely at 20°C unless precipitation or biological activity is observed in solution.
Glucose	196.0 mg	
Millipore water	500 mL	

9. Procedure

9.1 Sample Collection

- 9.1.1 Prepare a Van Dorn bottle to be deployed.
- 9.1.2 Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3 Drop the bottle messenger to close the Van Dorn at the desired depth.
- 9.1.4 Pull the Van Dorn bottle back up into the boat.
- 9.1.5 Connect the Tygon tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6 Push open the Van Dorn bottle's spigot valve to allow some water to flow out before proceeding.
- 9.1.7 Rinse the polypropylene collection bottle three times with water from the Van Dorn bottle.
- 9.1.8 Collect approximately 500 mL into the 1 L sample collection bottle.
- 9.1.9 Place the sample bottle inside the cooler until analysis in the lab.
 - 9.1.9.1 Keep the samples at 4°C from the time of collection and not more than 48 hours before beginning sample analysis.

9.2 Calibration

- 9.2.1 Fill a BOD bottle approximately $\frac{1}{3}$ of the total volume with freshwater to create water-saturated air and wait several minutes for the air to reach equilibrium with the water.
- 9.2.2 Turn on the Orion DO meter with BOD probe and press the "cal" button when in measurement mode.
- 9.2.3 Use the arrows to highlight the "Air" calibration option and "select."
- 9.2.4 Rinse the BOD probe with Millipore water and blot dry with a kimwipe.
- 9.2.5 Place the BOD probe into the BOD bottle $\frac{1}{3}$ filled with freshwater so the probe is in the air above the water when added to the BOD bottle.
- 9.2.6 Press "start" to begin the calibration.
- 9.2.7 Wait for the reading to stabilize and wait for the message "Accepting Auto % Sat. Calibration 102.3%"
- 9.2.8 Press "cal done".

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

9.2.9 Press “f1” to return to measurement mode to measure the water saturated air.

9.2.10 The BOD probe should read 8.5 ± 0.1 mg/L at 23.5°C for the 100% saturated air after calibration.

9.2.10.1 Check the BOD probe manual if the probe measurement does not agree with the expected value.

9.2.10.2 Record the calibration temperature and DO of the air saturated water after calibration in the BOD lab notebook.

9.2.10.3 If the temperature is different, you can calculate the expected BOD using USGS DOTABLES: <https://water.usgs.gov/cgi-bin/dotables>

9.3 Sample Analysis

9.3.1 Bring samples to $20 \pm 3^\circ\text{C}$.

9.3.2 Shake the sample bottle vigorously with fresh air to liberate any oxygen from photosynthesis and equilibrate the bottle with laboratory air.

9.3.2.1 Fill the BOD bottle halfway with the sample seawater.

9.3.2.2 Add 300 μL of each of the prepared phosphate buffer, MgSO_4 , CaCl_2 , and FeCl_3 solutions to the 300 mL BOD bottle with sample water and mix thoroughly.

9.3.2.3 Fill the BOD bottle to the base of the bottle neck with remaining seawater sample.

9.3.3 For determination of initial DO:

9.3.3.1 Calibrate the BOD sensor prior to each day of use (Section 9.2)

9.3.3.2 Rinse with DI water and blot dry the BOD probe with a Kimwipe.

9.3.3.3 Place the Orion AUTO-STIR BOD Probe directly into the BOD bottle.

9.3.3.4 The “stirrer” should turn off automatically. If not, press the “stirrer” button to engage the stirrer. The stirrer will engage and a green light will appear on top of the stirrer.

9.3.3.5 Press the “measure” button to measure the sample.

9.3.3.6 Wait for the screen to say “ready” and the “AR” symbol stops flashing.

9.3.3.7 Record the initial sample DO (mg/L) and temperature ($^\circ\text{C}$) in the BOD lab notebook.

9.3.3.8 The “stirrer” should turn off automatically. If not, press the “stirrer” button to turn off the stirrer. The stirrer will turn off and the green light will turn off on top of the stirrer.

9.3.3.9 Remove the probe from the BOD bottle, rinse with DI water, and blot dry with a Kimwipe.

9.3.4 Fill the BOD bottle to the base of the bottle neck with remaining seawater sample, tap stopper to remove bubbles, refill the neck of the bottle if needed, tilt bottle slightly, place stopper, and clean any residue on the outside of the bottle.

9.3.4.1 Note: The bottle should be completely filled with seawater and no gas headspace during the incubation. Add additional sample seawater if necessary to avoid gas headspace in the bottle.

9.3.5 Create a water seal by adding additional sample water to the flared bath of the sample bottle lid and cap with an aluminum foil hat to reduce evaporation.

9.3.6 Incubate the stoppered and sealed BOD bottles at $20^\circ\text{C} \pm 1^\circ\text{C}$ for 5 days (± 6 hours).

9.3.7 Determination of final DO.

9.3.7.1 After $5\text{d} \pm 6\text{ h}$ of incubation, remove the sample from the incubator.

9.3.7.2 Calibrate the DO sensor prior to each day of use (Section 9.2).

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

- 9.3.7.3 Rinse with DI water and blot dry the BOD probe with a KimWipe.
- 9.3.7.4 Place the Orion AUTO-STIR BOD Probe directly into the BOD bottle.
- 9.3.7.5 Press the “stirrer” button to engage the stirrer.
- 9.3.7.6 Wait for the screen to say “ready” and record the final sample DO (mg/L) and temperature (°C) in the BOD lab notebook.
- 9.3.7.7 Press the “stirrer” button to turn off the stirrer.
- 9.3.7.8 Remove the probe from the BOD bottle.
- 9.3.8 Calculate the sample BOD as per the following equation:

$$\text{BOD (mg/L)} = \text{DO}_{\text{initial}} \text{ (mg/L)} - \text{DO}_{\text{final}} \text{ (mg/L)}$$
 where:
 $\text{DO}_{\text{initial}}$ = Initial DO of BOD sample (mg/mL)
 DO_{final} = Final DO of BOD sample after incubation (mg/L)

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Calibration data and sample DO readings will be recorded in the BOD station lab notebook.
- 10.3. Field and Calibration forms shall be kept in the designed file cabinet at CCRI office.
- 10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.5. All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.
- 10.6. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. Follow these steps to clean the BOD probe (every 1-3 months):
 - 11.1.0. Unscrew the membrane cap from the probe and dispose of the electrolyte solution and membrane cap.
 - 11.1.1. Place a few drops of distilled water on a polishing disk. Polish the probe cathode with the polishing disk for about 10 seconds. Polishing disks are included in the probe maintenance kits, Cat No. 080513 and 080113.
 - 11.1.2. Rinse the probe anode/cathode assembly with distilled water and blot dry.
 - 11.1.3. Fill a new membrane cap about ¾ full with the new electrolyte solution.
 - 11.1.4. Tap the side of the membrane cap to release any bubbles from the electrolyte solution.
 - 11.1.5. Screw the membrane cap onto the probe until the membrane cap is hand tight.
 - 11.1.6. Polarize the probe (see “DO Probe Maintenance” in user manual).

12. Troubleshooting

- 12.1. Ensure the probe remains polarized. If the DO meter is disconnected from electricity for an extended period of time, it is possible that the probe will need to be polarized again.
 - 12.1.0. Leave the probe connected and on for 1 hour to polarize the DO probe.

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Biochemical Oxygen Demand	WQP 011	1.1	

- 12.2. Rinse the probe thoroughly with distilled water and follow the cleaning procedure in the DO Probe Maintenance section in the user's manual.
- 12.3. If readings continue to be erratic and unstable, you may need to replace the membrane cap, replenish electrolyte or polish the cathode with a polishing disk (see Section 11.1).

13. QA/QC

- 13.1 As per QAPP Table 4, seawater BOD should be measured in milligrams per liter (mg/L) with an accuracy of ± 0.2 mg/L and a precision of $< 20\%$.
 - 13.1.1 Precision should be assessed via analysis of lab duplicates for 10% of samples.
- 13.2 Add 1.0 mL Glucose Glutamic Acid Solution to a sample duplicate to generate a laboratory fortified matrix duplicate.
 - 13.2.1 The added BOD by the Glucose Glutamic Acid Solution should be 2.0 ± 0.8 mg/L.
 - 13.2.2 The difference between the laboratory fortified matrix duplicate and the sample should be within $\pm 20\%$ of the sample BOD plus 2.0 ± 0.8 mg/L.
 - 13.2.3 Laboratory fortified matrix duplicate should be assessed for 10% of samples.
 - 13.2.4 All solutions should be freshly prepared approximately monthly for each sampling event to avoid degradation of Glucose Glutamic Acid solution and/or contamination of other solutions by biological activity.
- 13.3 The lower reporting limit is the detection limit for the BOD probe measurement (± 0.2 mg/L).
- 13.4 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.
- 13.5 An Onset HOBO Temperature Logger will be deployed in the incubator and checked at the end of each measurement cycle to ensure temperatures were maintained throughout the incubation period.

14. References

- 14.1. U.S. ENVIRONMENTAL PROTECTION AGENCY. 1993. Methods for Determination of Inorganic Substances in Environmental Samples; EPA-600/R/93/100—Draft. Environmental Monitoring Systems Lab., Cincinnati, Ohio.
- 14.2. Thermo Scientific Orion Auto Star A213 benchtop RDO/DO meter User Manual.
- 14.3. Thermo Scientific Orion AUTO-STIR BOD Probe User Manual.